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CSA0911

JAVA

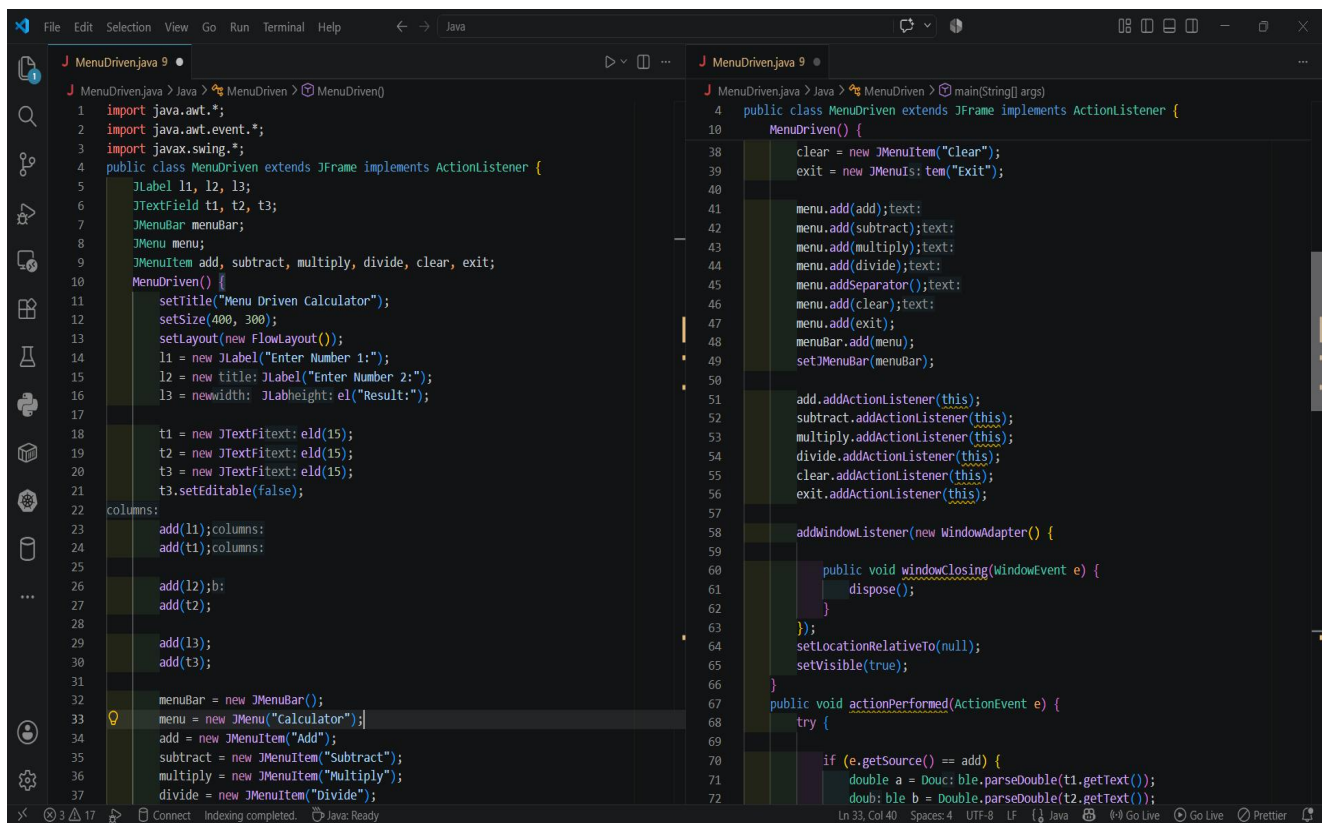
CO4 AT2 Q2

Q2 menu driven

1 mark

Create a menu-driven Swing application for a simple calculator (Add, Subtract, Multiply, Divide, Clear, Exit) using MenuBar, Menu, and MenuItem.

CODE:



```
1 import java.awt.*;
2 import java.awt.event.*;
3 import javax.swing.*;
4 public class MenuDriven extends JFrame implements ActionListener {
5     JLabel l1, l2, l3;
6     JTextField t1, t2, t3;
7     JMenuBar menuBar;
8     JMenu menu;
9     JMenuItem add, subtract, multiply, divide, clear, exit;
10    MenuDriven() {
11        setTitle("Menu Driven Calculator");
12        setSize(400, 300);
13        setLayout(new FlowLayout());
14        l1 = new JLabel("Enter Number 1:");
15        l2 = new JLabel("Enter Number 2:");
16        l3 = new JLabel("Result:");
17
18        t1 = new JTextField(15);
19        t2 = new JTextField(15);
20        t3 = new JTextField(15);
21        t3.setEditable(false);
22
23        add(l1);
24        add(t1);
25
26        add(l2);
27        add(t2);
28
29        add(l3);
30        add(t3);
31
32        menuBar = new JMenuBar();
33        menu = new JMenu("Calculator");
34        add = new JMenuItem("Add");
35        subtract = new JMenuItem("Subtract");
36        multiply = new JMenuItem("Multiply");
37        divide = new JMenuItem("Divide");
38        clear = new JMenuItem("Clear");
39        exit = new JMenuItem("Exit");
40
41        menu.add(add);
42        menu.add(subtract);
43        menu.add(multiply);
44        menu.add(divide);
45        menu.addSeparator();
46        menu.add(clear);
47        menu.add(exit);
48        menuBar.add(menu);
49        setJMenuBar(menuBar);
50
51        add.addActionListener(this);
52        subtract.addActionListener(this);
53        multiply.addActionListener(this);
54        divide.addActionListener(this);
55        clear.addActionListener(this);
56        exit.addActionListener(this);
57
58        addWindowListener(new WindowAdapter() {
59            public void windowClosing(WindowEvent e) {
60                dispose();
61            }
62        });
63        setLocationRelativeTo(null);
64        setVisible(true);
65
66        public void actionPerformed(ActionEvent e) {
67            try {
68                if (e.getSource() == add) {
69                    double a = Double.parseDouble(t1.getText());
70                    double b = Double.parseDouble(t2.getText());
71                    double result = a + b;
72                    t3.setText(String.valueOf(result));
73                }
74            } catch (NumberFormatException ex) {
75                t3.setText("Invalid input");
76            }
77        }
78    }
79}
```

The screenshot shows an IDE with two open Java files. The left file, `MenuDriven.java`, contains the `MenuDriven` class which implements `ActionListener`. It has methods for adding, subtracting, multiplying, and dividing two numbers, as well as a clear button. The right file, also named `MenuDriven.java`, contains the `main` method which creates a `MenuDriven` object and runs it. The code is as follows:

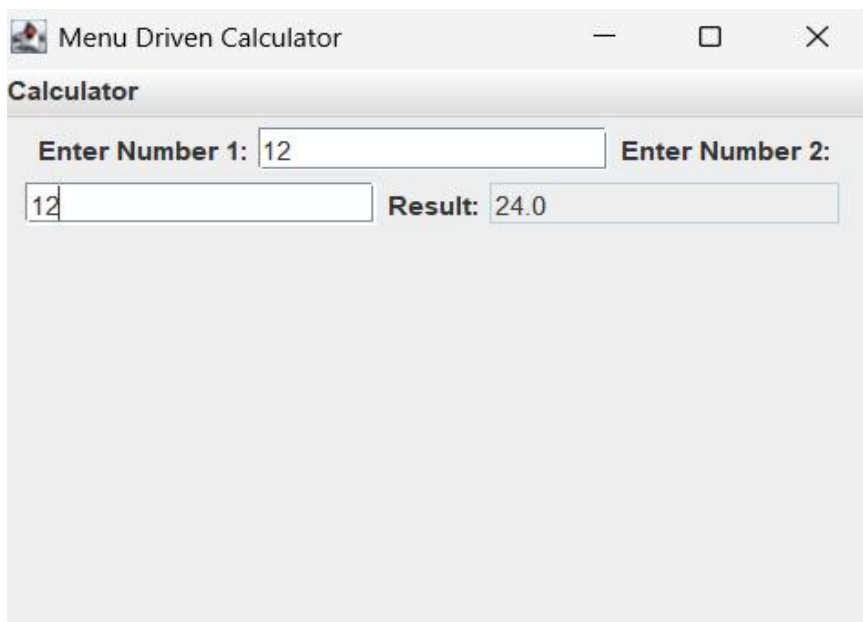
```
public class MenuDriven extends JFrame implements ActionListener {
    public void actionPerformed(ActionEvent e) {
        double a = Double.parseDouble(t1.getText());
        double b = Double.parseDouble(t2.getText());
        double c = Double.parseDouble(t3.getText());

        if (e.getSource() == add) {
            c = a + b;
        } else if (e.getSource() == subtract) {
            c = a - b;
        } else if (e.getSource() == multiply) {
            c = a * b;
        } else if (e.getSource() == divide) {
            if (b == 0) {
                t3.setText("Cannot divide by zero");
            } else {
                c = a / b;
            }
        } else if (e.getSource() == clear) {
            t1.setText("");
            t2.setText("");
            t3.setText("");
        }
        t3.setText(String.valueOf(c));
    }
}

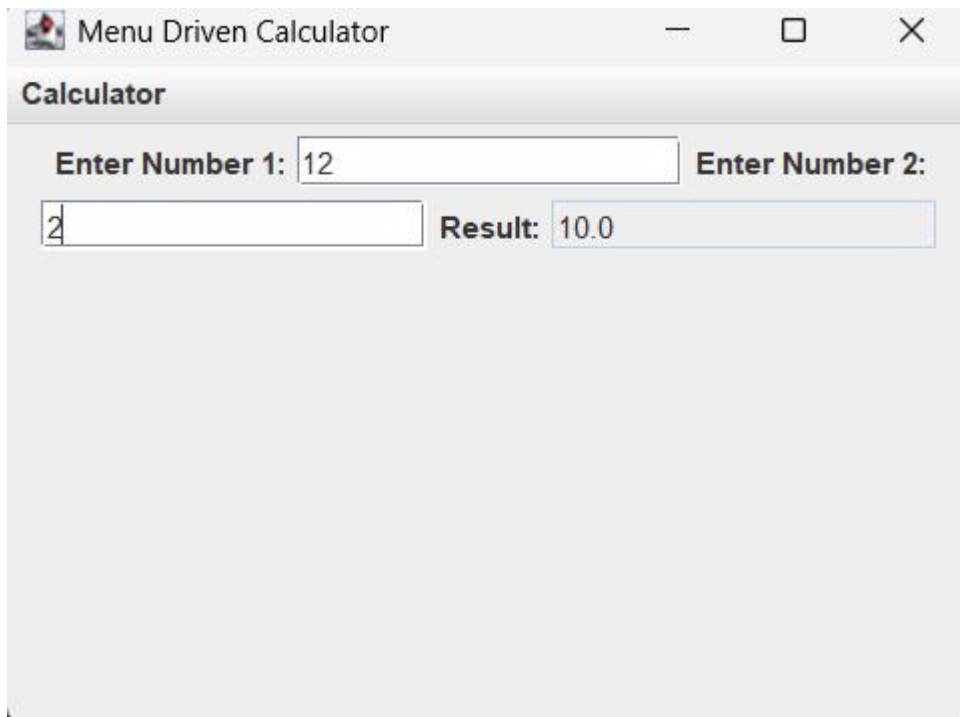
public static void main(String[] args) {
    new MenuDriven();
}
```

OUTPUT:

Addition:



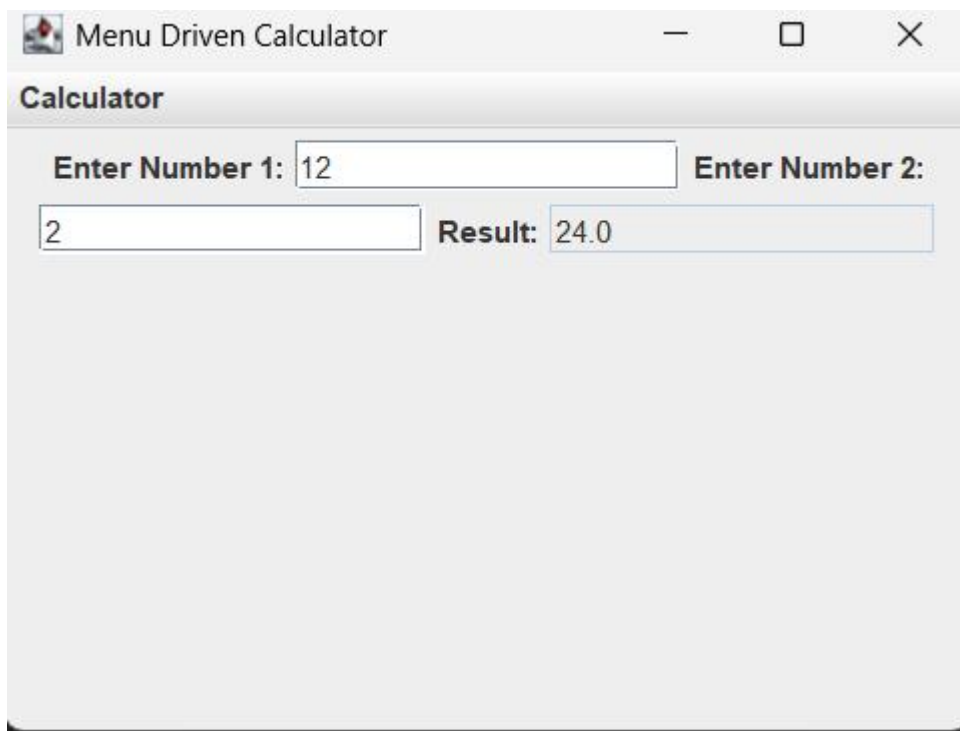
Subtraction:



The screenshot shows a window titled "Menu Driven Calculator" with a standard macOS-style title bar (minimize, maximize, close buttons). Below the title bar is a header bar labeled "Calculator". The main area contains two input fields: "Enter Number 1:" with the value "12" and "Enter Number 2:" which is empty. Below these, there is a third input field containing the digit "2" and a "Result:" field displaying "10.0".

Field	Value
Enter Number 1:	12
Enter Number 2:	
Result:	10.0

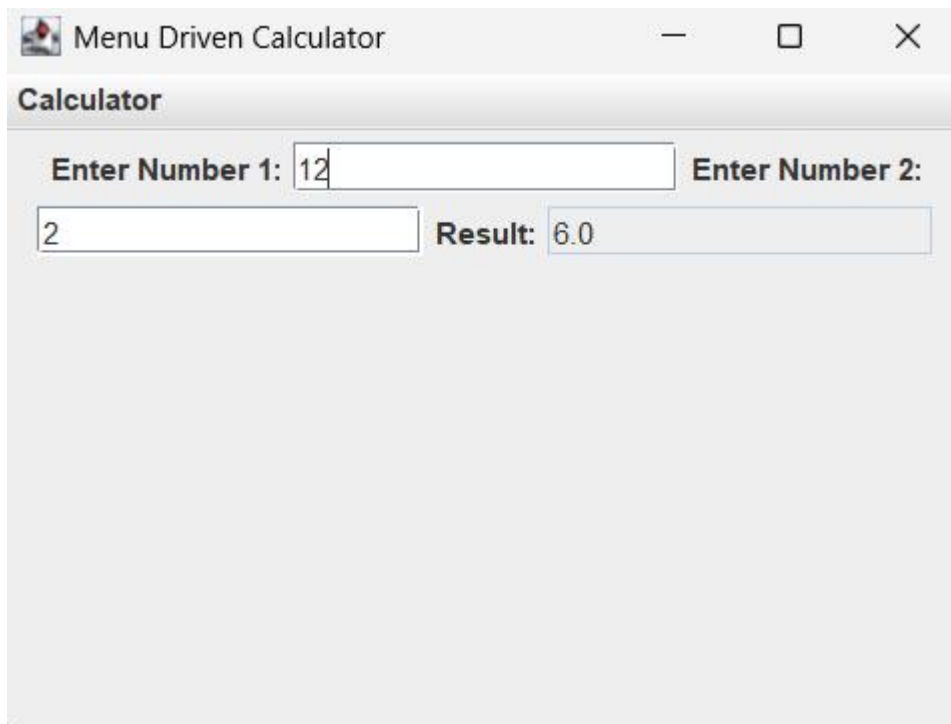
Multiplication:



The screenshot shows the same "Menu Driven Calculator" window. In this state, the "Enter Number 2:" field now contains the value "2", and the "Result:" field displays "24.0".

Field	Value
Enter Number 1:	12
Enter Number 2:	2
Result:	24.0

Division:



The image shows a window titled "Menu Driven Calculator" with a standard Windows-style title bar (minimize, maximize, close buttons). Below the title bar is a header bar labeled "Calculator". The main area of the window contains two input fields for numbers. The first input field is preceded by the text "Enter Number 1:" and contains the value "12". The second input field is preceded by the text "Enter Number 2:" and contains the value "2". Below these input fields, the text "Result:" is followed by a text box containing the value "6.0".

Input	Value
Enter Number 1:	12
Enter Number 2:	2
Result:	6.0